IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Agentic AI for Mental Health Awareness & Suicide Prevention

Domain of Project – Healthcare

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Problem Statement -

Problem Statement : Agentic AI for Mental Health Awareness & Suicide Prevention

The Challenge

Mental health issues such as depression, anxiety, and suicidal thoughts are growing worldwide. Stigma, lack of awareness, and limited access to timely help prevent many individuals from seeking support. Traditional systems are often reactive, leaving gaps in early detection and preventive care.

The Objective

Build an Agentic AI-powered Mental Health Awareness & Suicide Prevention Agent that is empathetic, proactive, and safe. The solution should:

1. **Promote Awareness & Education** – Share accessible resources on mindfulness, coping strategies, and self-care.
2. **Enable Early Detection (RAG)** – Use Retrieval-Augmented Generation to identify distress signals in conversations, journaling inputs, or safe online data.
3. **Provide Empathetic Support –** Build a conversational agent that listens actively and responds with empathy (with clear disclaimers: not a substitute for professional medical advice).
4. **Predict & Prevent Risks –** Forecast stress peaks, suggest preventive strategies, and issue alerts when high-risk signals are detected (with consent).
5. **Connect to Human Support –** Link users to helplines, counselors, or community support for timely intervention.

Technology Used

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- IBM watsonx.ai + Granite Models

I used IBM watsonx.ai , IBM Orchestrate + IBM Models

Proposed Solution -

Proposed Solution: AI-Powered Mental Health Awareness & Suicide Prevention Agent

The solution is a multi-agent AI system built using LangFlow, IBM watsonx.ai, **IBM watsonx Orchestrate,** and Granite Models to provide personalized mental health awareness and emotional support.

The system uses RAG (Retrieval-Augmented Generation) to retrieve trusted mental health information from sources such as WHO and mental health resources.

Key agents include:
- **Mental Health Knowledge Agent** – provides awareness and educational content
- **Emotional Support Agent** – offers supportive conversations and coping guidance
- **Risk Assessment Agent** – identifies emotional distress and encourages professional support
- **Mood Tracking Agent** – tracks mood and gives wellness feedback

IBM watsonx Orchestrate is used to coordinate and automate communication between multiple AI agents, manage conversation workflows, and deliver seamless user interactions.

Overall, the solution provides intelligent, real-time, and preventive mental health support through a unified AI platform.

Component Used -

1. **Chat Input**

Provides an interactive interface where users can enter their queries, emotions, concerns, or personal wellness details through **text, voice, or image-based input** to initiate communication with the system.

2. **IBM watsonx AI Agent**

Processes user inputs using **IBM watsonx.ai and Granite Models** to analyze emotional context and generate **personalized mental health awareness, emotional support, and wellness recommendations**.

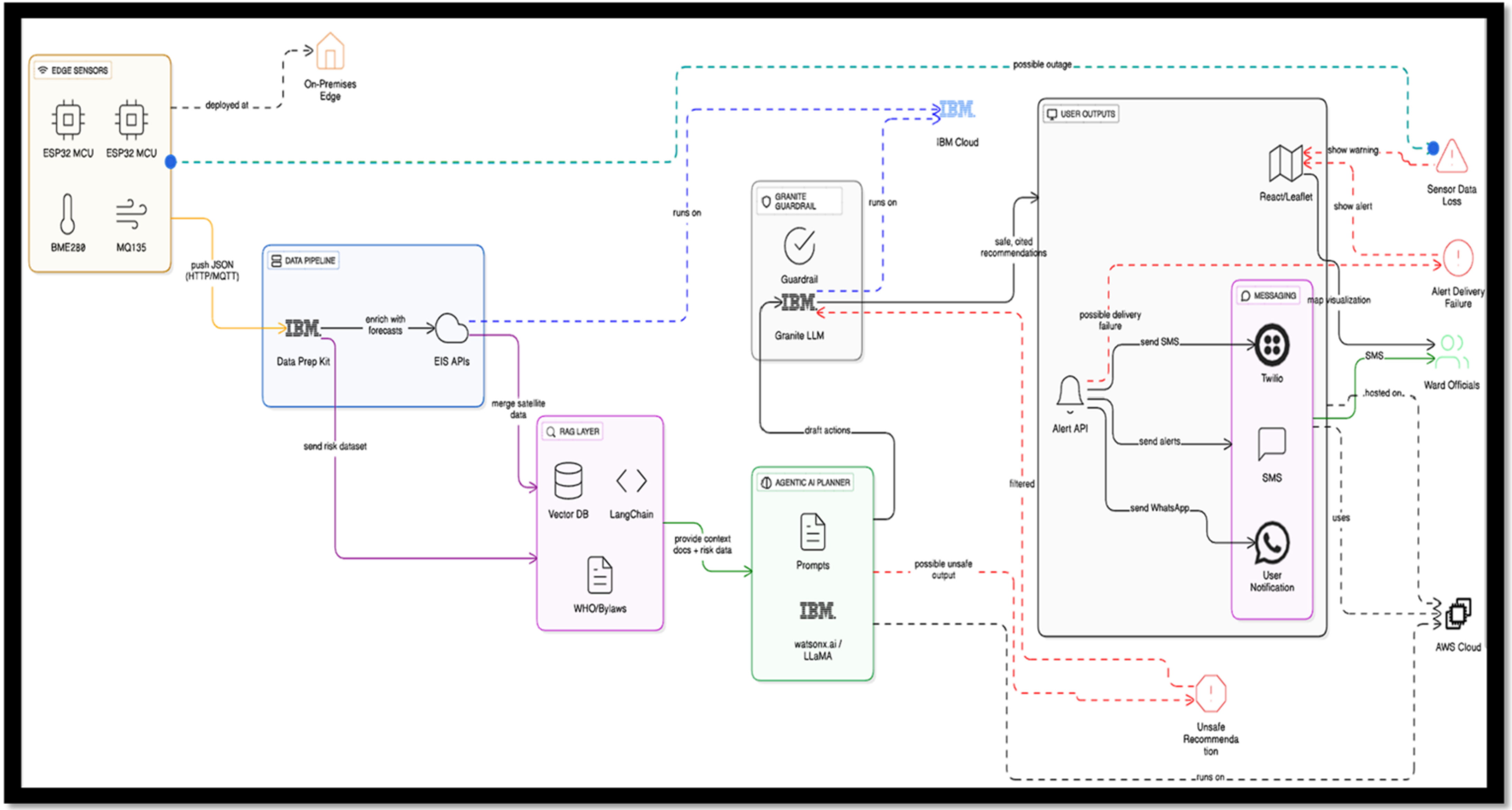
3. **Chat Output**

Displays AI-generated responses including **mental health guidance, emotional support messages, coping strategies, awareness information, and personalized recommendations**.

4. **File Component**

Stores and retrieves user interaction records, mood logs, wellness data, and awareness-related information for **progress tracking, analysis, and personalized support delivery**.

Architecture blueprint



Role of Agentic AI in the solution

Role of Agentic AI in Mental Health Awareness & Suicide Prevention

Agentic AI enables the Mental Health Awareness & Suicide Prevention platform to function as an intelligent, proactive support system rather than a traditional chatbot. Using **IBM watsonx.ai, IBM watsonx Orchestrate, and Granite Models**, the platform coordinates multiple specialized agents to provide personalized emotional support and mental health awareness.

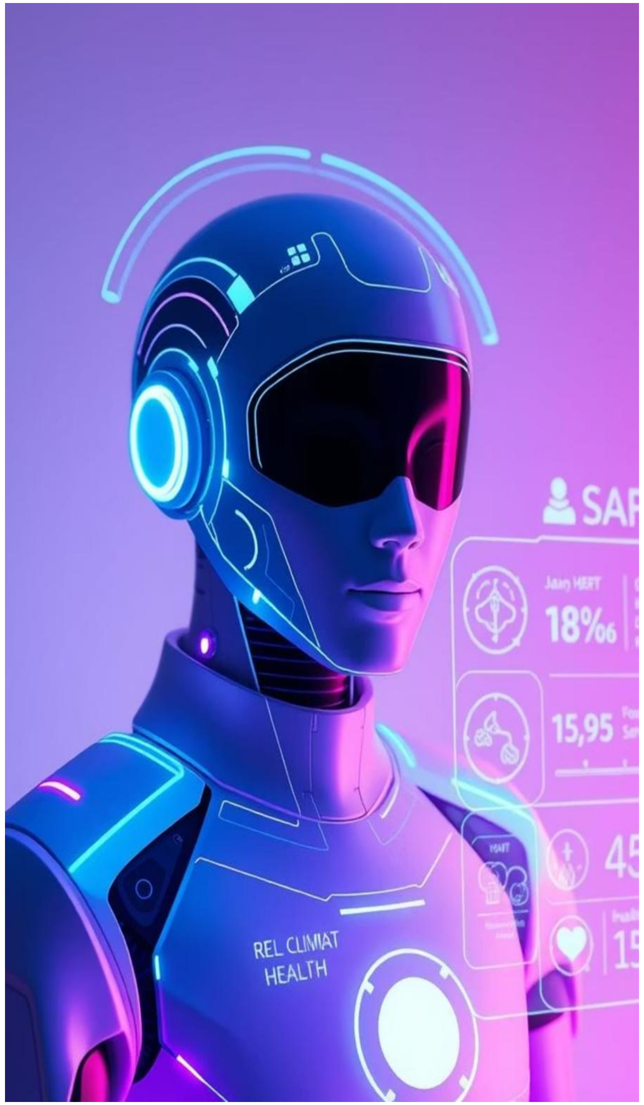
Each agent performs a dedicated role—such as **understanding user emotions, retrieving mental health knowledge, providing wellness recommendations, assessing emotional risk, and tracking user well-being**—while collaborating to deliver accurate and meaningful responses.

This multi-agent architecture improves context awareness by understanding user inputs such as emotional state, stress levels, concerns, and wellness goals to generate personalized guidance and support.

Agentic AI also enables real-time adaptation, continuously adjusting recommendations and responses based on ongoing conversations and user feedback.

Additionally, agents communicate and coordinate with each other through IBM watsonx Orchestrate, improving decision-making, workflow management, and response quality.

Overall, Agentic AI makes the system proactive, personalized, context-aware, and goal-driven, acting as a smart digital mental wellness assistant that promotes emotional well-being, early awareness, preventive care, and supportive intervention.



Technology Used

- **IBM Granite Model** (e.g., ibm-granite-3-2-8b): natural language understanding, emotional context analysis, reasoning, and personalized response generation. The model enables supportive conversations, mental health awareness guidance, coping recommendations, and contextual user interaction.

- Provides the **secure, reliable, and scalable infrastructure** required to build, deploy, and manage the Mental Health Awareness & Suicide Prevention platform efficiently.

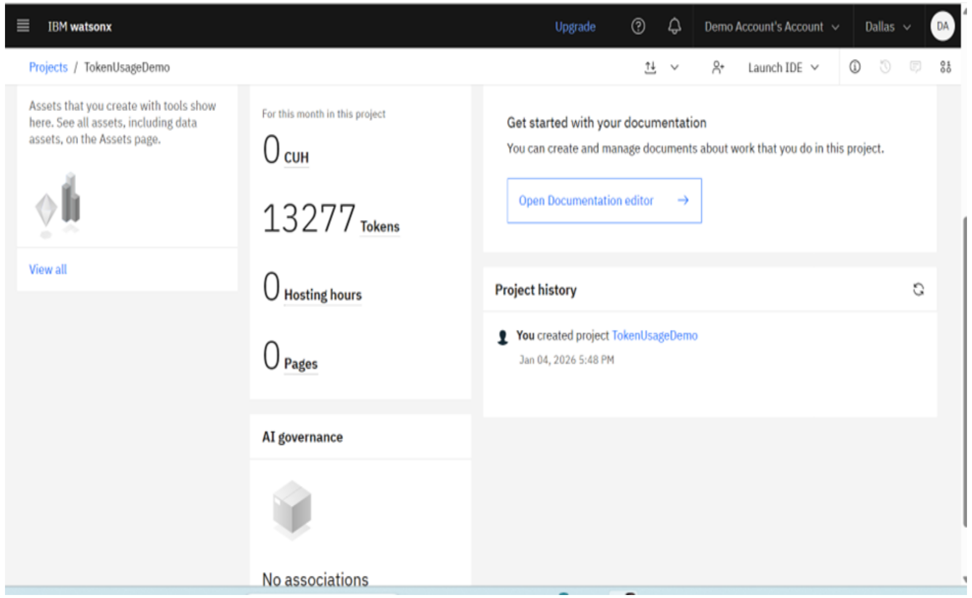
- **RAG** is used to retrieve trusted mental health information from verified knowledge sources before generating responses, ensuring that awareness content, emotional support guidance, and recommendations are more accurate, relevant, and trustworthy.

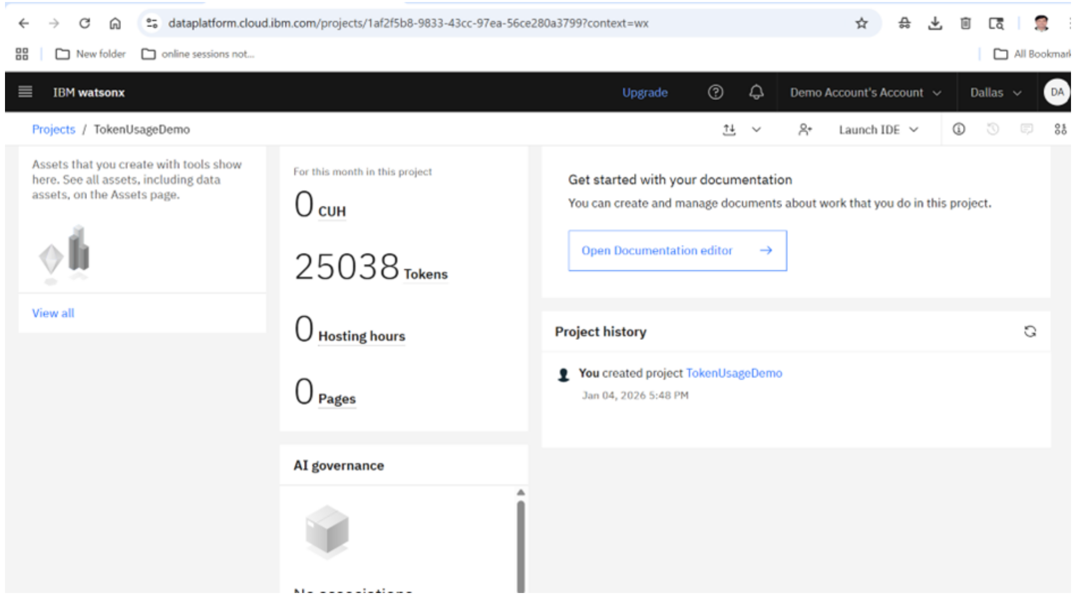
- **IBM watsonx.ai:** foundation model deployment, embeddings, prompt orchestration, AI governance, and responsible AI implementation** to support intelligent and scalable mental wellness interactions.

- **Vector Database (FAISS / Chroma):** Stores and retrieves embedded mental health awareness knowledge, emotional wellness resources, and support information for **fast and accurate RAG-based retrieval.**

Token Usage

11,761 token are used





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Novelty and Uniqueness

• **Multi-Agent AI** – Multiple intelligent agents collaborate for emotional support, awareness, and wellness guidance.
• **RAG-Based Responses** – Uses trusted mental health knowledge sources for accurate and reliable recommendations.
• **Personalized Support** – Adapts responses based on emotions, stress levels, and user context.
• **Real-Time Interaction** – Continuously updates guidance from ongoing user conversations.
• **Multi-Modal Input** – Supports text, voice, and image-based interaction.
• **Preventive Healthcare** – Focuses on early awareness and emotional well-being.
• **IBM watsonx Orchestrate** – Enables intelligent workflow coordination and seamless agent communication. **Acts as a proactive digital mental wellness assistant, not just a chatbot.**

Git Hub Link

Github URL – https://github.com/pravinkamble26/MentalhealthAgent.git

Uploaded following three files into github repository

1) app.json file , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4)any other files

Future Scope

⯁ Wearable Device Integration
Monitor sleep, stress, and wellness indicators for personalized support.

⯁ AI Voice Assistant & Multilingual Support
Enable natural voice interaction and support multiple languages.

⯁ Early Emotional Distress Detection
Use AI to identify emotional changes and provide timely wellness guidance.

⯁ Professional Support Integration
Connect users with mental health resources and support services.

⯁ Personalized Wellness Dashboard
Track mood, emotional trends, and well-being progress.

Thank You!

Thank you for your time and interest.